Unit 08 Practice Problems: Hypothesis Testing

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Required Practice Problems

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and/or preparing for exams, make sure you have completed and reviewed all practice problems from the unit. The **solutions** are posted on Canvas (`Unit-PP-08-Solution.pdf > Practice Problem-Solution > STAT 312 > Files`).

8.1

Phil is driving his car and is halfway to Jake's house when he realizes he may have forgotten the homemade blueberry pie he promised to bring. If he decides he forgot the pie, he will turn around and go back home to look for it. If he decides the pie is somewhere in his car, he will continue driving to Jake's house. Phil is faced with the following hypothesis test:

$$H_0 : \text{Phil forgot the pie,}$$
$$H_a : \text{Phil did not forget the pie.}$$

Describe each of the following situations as a Type I error, a Type II error, or no error.

(a) Phil goes back home to look for the pie, but he did not forget it after all. It was in his car the whole time.

(b) Phil goes back home, finds the pie, and then brings it to Jake's house.

(c) Phil continues driving to Jake's house and shows up without a pie.

(d) Phil continues driving to Jake's house and shows up with a pie. He did not forget it after all.

8.2

Jennifer gives you a coin and tells you it is either a fair coin with $P(H) = 0.50$ or an unfair coin with $P(H) = 0.80$, where H denotes heads. You will conduct the following hypothesis test:

$$H_0 : P(H) = 0.50 \qquad \text{versus} \qquad H_a : P(H) = 0.80.$$

You will flip the coin four times and reject H_0 if and only if you observe 3 or 4 heads.

(a) Find $\alpha = P(\text{Type I error})$.

(b) Find $\beta = P(\text{Type II error})$ and the power of the test.

8.3

The null hypothesis, alternative hypothesis, and significance level for three hypothesis tests are given below. All three are one-sample hypothesis tests for μ with known σ. Draw pictures of the rejection regions.

(a) $H_0 : \mu = 7, \quad H_a : \mu \neq 7, \quad \alpha = 0.0394.$

(b) $H_0 : \mu = 21.5, \quad H_a : \mu > 21.5, \quad \alpha = 0.0301.$

(c) $H_0 : \mu = 9.004, \quad H_a : \mu < 9.004, \quad \alpha = 0.0166.$

8.4

According to a wildlife survey conducted in Antarctica ten years ago, the mean height among all adult emperor penguins was 110 cm. A biologist named Sarah wants to know whether the mean height is different now. According to new data collected last month, a random sample of 32 adult emperor penguins had a sample mean height of 110.6 cm. The population standard deviation is known to be 4.5 cm. Conduct all steps of a hypothesis test with $\alpha = 0.0784$. What assumptions, if any, are required to conduct this hypothesis test?

8.5

Ceramic bars are designed to have a population mean fracture strength of 90 MPa. However, engineers studying the ceramic bars suspect that the population mean is actually less than 90 MPa. A random sample of 20 ceramic bars has a sample mean fracture strength of 85.32 MPa, and the population standard deviation is known to be 10.3 MPa. Conduct all steps of a hypothesis test with $\alpha = 0.0465$. What assumptions, if any, are required to conduct this hypothesis test?

8.6

A social scientist named Ashley reads that in 1960, the mean age among American men at first marriage was 25.3 years. She wants to know whether the mean age is greater now. Ashley does not know μ or σ. She takes a random sample of 20 American men who got married for the first time last month and records their ages on their wedding days. The sample mean is 29.1 years, and the sample standard deviation is 6.15 years. Conduct all steps of a hypothesis test with $\alpha = 0.025$. What assumptions, if any, are required to conduct this hypothesis test?

8.7

Over the course of several weeks, Mitchell has trained his lab rats to run through a maze. At the end of the maze, the rats are rewarded with breakfast. It has been established that his population of rats completes the maze in an average of 10.00 seconds. Mitchell wonders whether the time would be significantly less if he had the rats run the maze at a different time of day. As an experiment, he randomly selects 17 of the rats and has them run the maze immediately after he feeds them dinner. The sample results are $n = 17$, $\overline{X} = 9.52$ seconds, and $s = 1.4$ seconds; σ is unknown. Conduct all steps of a hypothesis test with $\alpha = 0.05$. What assumptions, if any, are required to conduct this hypothesis test?

8.8

The president of Big State University claims that 75% of all students at the university are registered to vote. You suspect that the true proportion is less than 0.75 and take a random sample. Of 80 randomly selected students, 56 are registered to vote. Conduct all steps of a hypothesis test with $\alpha = 0.01$.

8.9

Many years ago, the proportion of citizens of Japan who were over 65 years old was 0.16. Demographers would like to know whether the proportion is greater now. A random sample of $n = 120$ Japanese citizens yielded 30 who were over 65. Conduct all steps of a hypothesis test with $\alpha = 0.025$.

8.10

A random sample of size $n = 49$ will be taken from a normal population with unknown mean μ and known standard deviation $\sigma = 14$. Daisy will conduct a one-sample Z-test:

$$H_0 : \mu = 61 \qquad \text{versus} \qquad H_a : \mu \neq 61.$$

(a) If $\alpha = 0.1212$, draw a picture of the rejection region on the appropriate curve and identify the critical value(s).

(b) If the actual value of μ equals 62, find the probability of a Type II error and the power of the test.

8.11

A random sample of size $n = 16$ will be taken from a normal population with unknown mean μ and known standard deviation $\sigma = 28$. Samantha will conduct a one-sample Z-test:

$$H_0 : \mu = 90 \qquad \text{versus} \qquad H_a : \mu < 90.$$

(a) If $\alpha = 0.0274$, draw a picture of the rejection region on the appropriate curve and identify the critical value(s).

(b) If the actual value of μ equals 85, find the probability of a Type II error and the power of the test. If the sample size is increased while keeping the significance level fixed, what will happen to these probabilities? Repeat the calculations with $n = 196$.

Supplemental Practice Problems

The problems below provide additional practice opportunities for students who would like more practice with the concepts covered in this unit.
Chapter 9 (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger)

- 9.1.1, 9.1.3, 9.1.7, 9.2.1 (b), 9.2.2, 9.2.6a, 9.3.1, 9.3.3, 9.3.5, 9.3.6a, 9.4.1, 9.4.4, 9.5.1, and 9.5.4

[To find these exercises in the digital copy of the textbook (`Textbooks > STAT 312 > Files > Canvas`), go to the xvi Contents page and click the "EXERCISES" tab located at the lower-right corner of the page. This will open the complete list of chapter exercises. The **solutions** are posted on Canvas (`textbook-solution.pdf > Practice Problem-Solution > STAT 312 > Files`).]